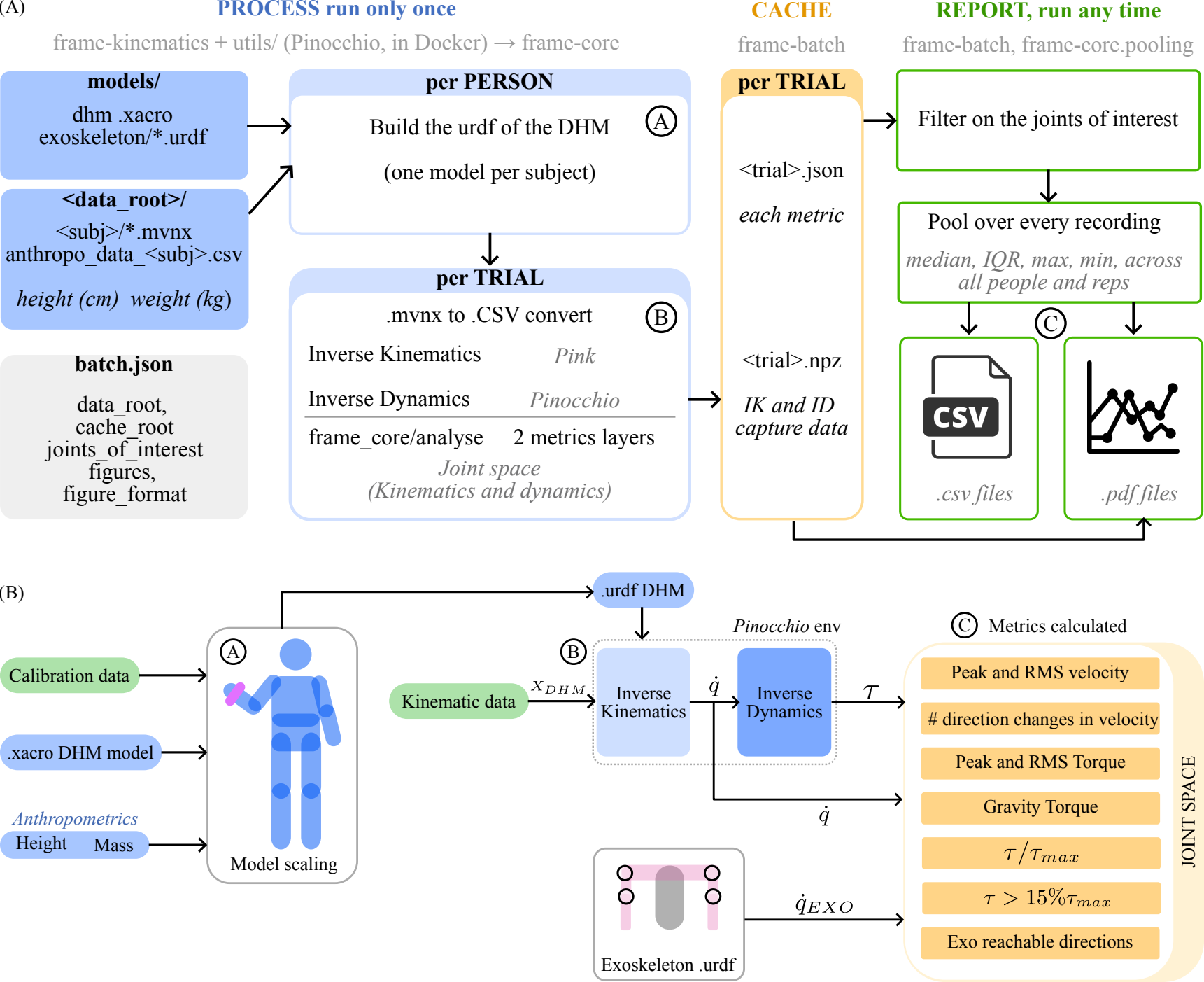

(A)
PROCESS run only once
frame-kinematics + utils/ (Pinocchio, in Docker) → frame-core
CACHE
frame-batch
REPORT, run any time
frame-batch, frame-core.pooling
models/
dhm .xacro
exoskeleton/*.urdf
<data_root>/
<subj>/*.mvnx
anthropo_data_<subj>.csv
height (cm) weight (kg)
batch.json
data_root,
cache_root
joints_of_interest
figures,
figure_format
per PERSON
Build the urdf of the DHM
A
(one model per subject)
per TRIAL
.mvnx to .CSV convert
B
Inverse Kinematics
Pink
Inverse Dynamics
Pinocchio
frame_core/analyse
2 metrics layers
Joint space
(Kinematics and dynamics)
per TRIAL
<trial>.json
each metric
<trial>.npz
IK and ID
capture data
Filter on the joints of interest
Pool over every recording
median, IQR, max, min, across
all people and reps
C
CSV
.csv files
.pdf files
(B)
.urdf DHM
Calibration data
.xacro DHM model
Anthropometrics
Height
Mass
A
Model scaling
Kinematic data
x_{DHM}
B
Pinocchio env
Inverse
Kinematics
$\dot{q}$
Inverse
Dynamics
τ
$\dot{q}$
Exoskeleton .urdf
$\dot{q}_{EXO}$
C
Metrics calculated
Peak and RMS velocity
direction changes in velocity
Peak and RMS Torque
Gravity Torque
τ/τ_{max}
$\tau > 15\%\tau_{max}$
Exo reachable directions
JOINT SPACE